Week #9: Interpolation and Approximation

Numerical Methods: Mathematical Foundations

June 22, 2025

Overview

Interpolation and approximation form fundamental tools in numerical analysis, providing methods to construct functions that approximate given data or known functions. This week we develop the mathematical theory underlying polynomial interpolation, spline methods, and approximation techniques, analyzing their convergence properties and practical applications.

The central challenge in interpolation is to construct a function that passes through given data points while maintaining desirable mathematical properties such as smoothness and numerical stability. We explore how different interpolation methods balance these competing requirements and examine the trade-offs between accuracy and computational efficiency.

Our analysis emphasizes the mathematical foundations that determine approximation quality, the role of basis functions in constructing interpolants, and the error analysis that guides method selection and parameter tuning.

1 Polynomial Interpolation

1.1 Existence and Uniqueness

Theorem 1.1 (Fundamental Theorem of Polynomial Interpolation)**.** *Given $n+1$ distinct points $(x_0, y_0), (x_1, y_1), \ldots, (x_n$ there exists a unique polynomial $p(x)$ of degree at most n such that $p(x_i) = y_i$ for $i = 0, 1, \ldots, n$.*

Proof. The interpolation conditions form a linear system $Vc = y$ where V is the Vandermonde matrix with $V_{ij} = x_i^j$. Since the points are distinct, V is nonsingular, ensuring existence and uniqueness. □

1.2 Lagrange Interpolation

Definition. Lagrange Interpolating Polynomial

The Lagrange form of the interpolating polynomial is:

$$p(x) = \sum_{i=0}^{n} y_i L_i(x)$$

where the Lagrange basis polynomials are:

$$L_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^{n} \frac{x - x_j}{x_i - x_j}$$

The Lagrange basis has the property $L_i(x_j) = \delta_{ij}$ (Kronecker delta).

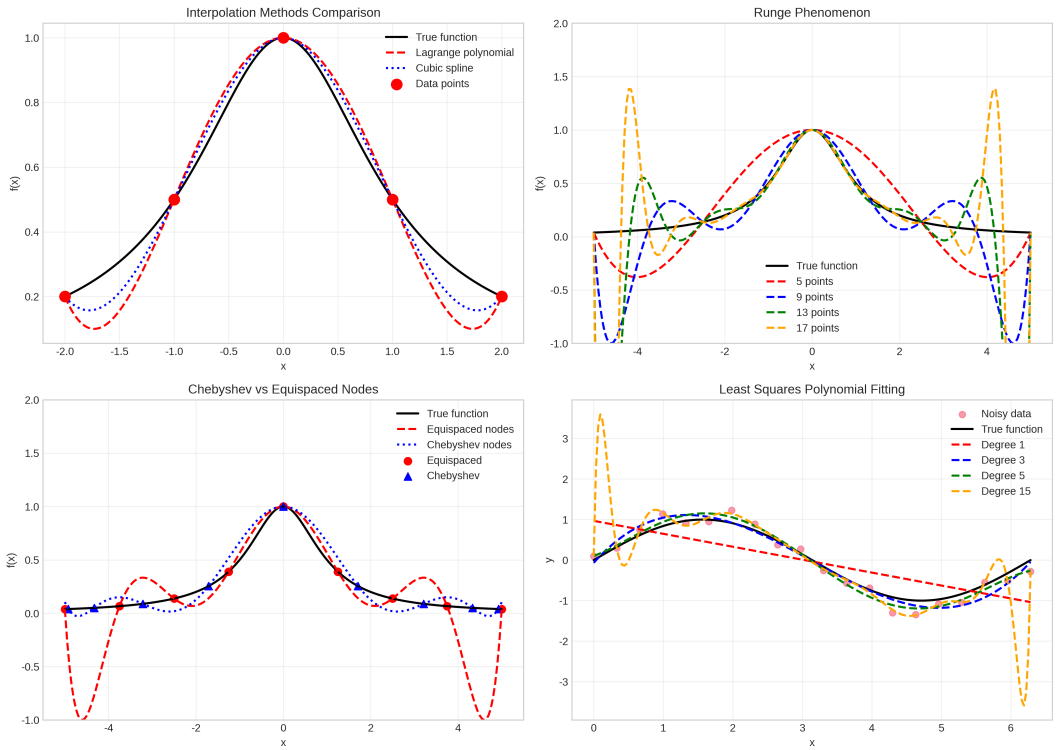

Figure 1: Interpolation methods comparison. The left panel shows polynomial interpolation with different node distributions. The right panel illustrates spline interpolation and its superior behavior near boundaries compared to high-degree polynomials.

1.3 Newton Interpolation

Definition. Divided Differences

The divided differences are defined recursively:

$$f[x_i] = f(x_i) \tag{1}$$

$$f[x_i, x_{i+1}, \ldots, x_{i+k}] = \frac{f[x_{i+1}, \ldots, x_{i+k}] - f[x_i, \ldots, x_{i+k-1}]}{x_{i+k} - x_i} \tag{2}$$

Definition. Newton Interpolating Polynomial

The Newton form is:

$$p(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) + \cdots$$

The Newton form is particularly useful for adding new interpolation points incrementally.

2 Interpolation Error Analysis

2.1 Error Formula

Theorem 2.1 (Interpolation Error Formula)**.** *If $f \in C^{n+1}[a, b]$ and p is the polynomial interpolating f at points $x_0, \ldots, x_n \in [a, b]$, then for any $x \in [a, b]$:*

$$f(x) - p(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^{n} (x - x_i)$$

for some $\xi \in [a, b]$.

Proof. For fixed x, define $g(t) = f(t) - p(t) - K \prod_{i=0}^{n}(t - x_i)$ where K is chosen so that $g(x) = 0$. Apply Rolle's theorem repeatedly to show that $g^{(n+1)}$ has a zero, leading to the error formula. □

2.2 Runge Phenomenon

High-degree polynomial interpolation with equally spaced points can exhibit severe oscillations.

Theorem 2.2 (Runge Phenomenon)**.** *For the function* $f(x) = 1/(1 + 25x^2)$ *on* $[-1, 1]$*, polynomial interpolation at equally spaced points diverges as the degree increases, with maximum errors growing exponentially.*

This motivates the use of Chebyshev points for better interpolation behavior.

2.3 Chebyshev Points

Definition. Chebyshev Points

The Chebyshev points on $[-1, 1]$ are:

$$x_k = \cos\left(\frac{(2k+1)\pi}{2(n+1)}\right), \quad k = 0, 1, \ldots, n$$

Theorem 2.3 (Optimal Interpolation Points)**.** *Among all possible choices of interpolation points, Chebyshev points minimize the maximum value of* $\prod_{i=0}^{n} |x - x_i|$ *on* $[-1, 1]$.

3 Spline Interpolation

Splines provide an alternative to high-degree polynomial interpolation by using piecewise polynomials.

3.1 Cubic Splines

Definition. Cubic Spline

A cubic spline $S(x)$ on the interval $[a, b]$ with knots $a = x_0 < x_1 < \cdots < x_n = b$ is a function such that:

1. $S(x)$ is a cubic polynomial on each subinterval $[x_i, x_{i+1}]$
2. $S \in C^2[a, b]$ (twice continuously differentiable)
3. $S(x_i) = y_i$ for $i = 0, 1, \ldots, n$

3.2 Construction of Cubic Splines

Let $S_i(x)$ be the cubic polynomial on $[x_i, x_{i+1}]$. The continuity conditions give:

$$S_i(x_i) = y_i, \quad S_i(x_{i+1}) = y_{i+1} \tag{3}$$
$$S_i'(x_{i+1}) = S_{i+1}'(x_{i+1}) \tag{4}$$
$$S_i''(x_{i+1}) = S_{i+1}''(x_{i+1}) \tag{5}$$

Theorem 3.1 (Natural Cubic Spline)**.** *The natural cubic spline (with* $S''(x_0) = S''(x_n) = 0$*) is uniquely determined and can be computed by solving a tridiagonal linear system.*

3.3 Spline Error Analysis

Theorem 3.2 (Cubic Spline Error Bound)**.** *If $f \in C^4[a,b]$ and S is the natural cubic spline interpolating f, then:*

$$\|f - S\|_\infty \leq \frac{5}{384} h^4 \|f^{(4)}\|_\infty$$

where $h = \max_i(x_{i+1} - x_i)$ is the maximum spacing.

This shows that splines achieve $O(h^4)$ convergence, much better than the $O(h^{n+1})$ of degree-n polynomial interpolation.

4 Least Squares Approximation

When data contains noise or we have more data points than desired polynomial degree, least squares approximation is appropriate.

4.1 Discrete Least Squares

Definition. Discrete Least Squares Problem

Given data points (x_i, y_i) for $i = 1, \ldots, m$ and basis functions $\phi_0, \phi_1, \ldots, \phi_n$ with $m > n$, find coefficients c_j to minimize:

$$\sum_{i=1}^{m} \left(y_i - \sum_{j=0}^{n} c_j \phi_j(x_i) \right)^2$$

This leads to the normal equations:

$$\sum_{j=0}^{n} c_j \sum_{i=1}^{m} \phi_j(x_i)\phi_k(x_i) = \sum_{i=1}^{m} y_i \phi_k(x_i), \quad k = 0, \ldots, n$$

4.2 Orthogonal Polynomials

Using orthogonal polynomials as basis functions improves numerical stability.

Definition. Orthogonal Polynomials

Polynomials $\{p_n(x)\}$ are orthogonal with respect to weight function $w(x)$ on $[a,b]$ if:

$$\int_a^b p_m(x) p_n(x) w(x) dx = 0 \quad \text{for } m \neq n$$

Common orthogonal polynomials include:

- **Legendre polynomials**: $w(x) = 1$ on $[-1, 1]$
- **Chebyshev polynomials**: $w(x) = 1/\sqrt{1 - x^2}$ on $[-1, 1]$
- **Hermite polynomials**: $w(x) = e^{-x^2}$ on $(-\infty, \infty)$

5 Trigonometric Interpolation

For periodic functions, trigonometric interpolation is often more appropriate than polynomial interpolation.

5.1 Discrete Fourier Transform

Definition. Trigonometric Interpolation

For $2n$ equally spaced points $x_j = j\pi/n$ with $j = 0, 1, \ldots, 2n-1$, the trigonometric interpolating polynomial is:

$$p(x) = \frac{a_0}{2} + \sum_{k=1}^{n-1}(a_k \cos(kx) + b_k \sin(kx)) + \frac{a_n}{2}\cos(nx)$$

The coefficients are given by the Discrete Fourier Transform (DFT):

$$a_k = \frac{1}{n}\sum_{j=0}^{2n-1} y_j \cos(kx_j) \tag{6}$$

$$b_k = \frac{1}{n}\sum_{j=0}^{2n-1} y_j \sin(kx_j) \tag{7}$$

5.2 Fast Fourier Transform

The Fast Fourier Transform (FFT) computes the DFT in $O(n \log n)$ operations instead of $O(n^2)$.

Theorem 5.1 (FFT Complexity)**.** *The FFT algorithm reduces the computational complexity of the DFT from $O(n^2)$ to $O(n \log n)$ by exploiting the symmetry properties of the roots of unity.*

6 Rational Interpolation

Rational functions can provide better approximations than polynomials for functions with poles or asymptotic behavior.

6.1 Padé Approximation

Definition. Padé Approximant

The (m, n) Padé approximant to function $f(x)$ is the rational function:

$$R_{m,n}(x) = \frac{P_m(x)}{Q_n(x)}$$

where P_m has degree at most m, Q_n has degree at most n, and the Taylor series of $R_{m,n}$ matches that of f through order $m + n$.

Padé approximants often converge faster than Taylor polynomials, especially for functions with singularities.

Intuition: Why Rational Approximation Works

Rational functions can model both polynomial behavior and asymptotic behavior (through poles). They provide a more flexible framework than polynomials alone, often achieving better approximation with fewer parameters.

7 Multivariate Interpolation

Extending interpolation to multiple dimensions presents new challenges.

7.1 Tensor Product Interpolation

Definition. Tensor Product Interpolation

For functions of two variables, tensor product interpolation uses:

$$p(x, y) = \sum_{i=0}^{m} \sum_{j=0}^{n} c_{ij} \phi_i(x) \psi_j(y)$$

where ϕ_i and ψ_j are one-dimensional basis functions.

7.2 Radial Basis Functions

Definition. Radial Basis Function Interpolation

RBF interpolation uses the form:

$$s(x) = \sum_{i=1}^{n} \lambda_i \phi(\|x - x_i\|)$$

where ϕ is a radially symmetric function and λ_i are coefficients determined by interpolation conditions.

Common RBF choices include:

- Gaussian: $\phi(r) = e^{-(\epsilon r)^2}$
- Multiquadric: $\phi(r) = \sqrt{1 + (\epsilon r)^2}$
- Thin plate spline: $\phi(r) = r^2 \log r$

Practice Problems

1. Prove that the Lagrange basis polynomials $L_i(x)$ form a partition of unity: $\sum_{i=0}^{n} L_i(x) = 1$.
2. Show that for Chebyshev points on $[-1, 1]$, the Lebesgue constant grows like $O(\log n)$ rather than exponentially.
3. Implement natural cubic spline interpolation and verify the $O(h^4)$ convergence rate.
4. Compare polynomial and rational interpolation for the function $f(x) = 1/(1 + x^2)$ on $[-5, 5]$.
5. Analyze the conditioning of the Vandermonde matrix and explain why it becomes ill-conditioned for large n.

Solution:

Problem 1: Consider the polynomial $q(x) = \sum_{i=0}^{n} L_i(x) - 1$. This polynomial has degree at most n and vanishes at the $n + 1$ distinct points $x_0, \ldots, x_n$. Therefore $q(x) \equiv 0$, proving the partition of unity property.

Problem 2: The Lebesgue constant for Chebyshev points is $\Lambda_n = O(\log n)$ because the Chebyshev points cluster near the endpoints, balancing the growth of the interpolation error across the interval.

Problem 3: Natural cubic splines achieve $O(h^4)$ convergence because they minimize the integrated squared second derivative while satisfying interpolation conditions, leading to optimal smoothness properties.

Problem 4: For $f(x) = 1/(1 + x^2)$, rational interpolation performs much better than polynomial interpolation because it can model the function's poles at $x = \pm i$ in the complex plane.

Problem 5: The Vandermonde matrix has condition number that grows exponentially with

n for equally spaced points. This is because the monomials $1, x, x^2, \ldots, x^n$ become nearly linearly dependent as n increases.